Railtrack Company Code of Practice

METHODOLOGY FOR THE DEMONSTRATION OF ELECTRICAL COMPATIBILITY BETWEEN ROLLING STOCK AND INFRASTRUCTURE

ENDORSEMENT & AUTHORISATION

Endorsed by:

....

A Simmons, Professional Head of Signal Engineering

Accepted for issue by:

....

P McAleer, Acting Head of Business Standards

Authorised by:

....

A McNaughton, Chief Engineer

ISSUE RECORD

Issue	Date	Comments
1	February 2003	New Code of Practice

IMPLEMENTATION

The provisions of this Code of Practice may be used from date of issue.

DISCLAIMER

Railtrack PLC, part of the Network Rail group of companies, has used its best endeavours to ensure that the content, layout and text of this document are accurate, complete and suitable for its stated purpose. It makes no warranties, express or implied, that compliance with the contents of this document shall be sufficient to ensure safe systems of work or operation. Railtrack PLC, part of the Network Rail group of companies, will not be liable to pay compensation in respect of the content or subsequent use of this document for any purpose other than its stated purpose or for any purpose other than that for which it was prepared except where it can be shown to have acted in bad faith or there has been wilful default.

SUPPLY

Copies of documents are available electronically, within Railtrack's organisation. Hard copies of this document will be available to Railtrack staff on request to the Document Controller and to other organisations from Technical Indexes Ltd (01344 404409).

Published & Issued by **Railtrack PLC (part of the Network Rail group of companies)**
Railtrack House
Euston Square
LONDON
NW1 2EE

Contents

1 PURPOSE 4

2 SCOPE 4

3 DEFINITIONS & ABBREVIATIONS 4

4 PRINCIPLE 5

5 PROCESS 5

5.1 IDENTIFICATION OF EQUIPMENT ON THE ROUTE 5
5.2 OBTAINING THE SUSCEPTIBILITY CHARACTERISTICS 6
5.3 COMPARISON OF TRAIN INTERFERENCE WITH SUSCEPTIBILITY LIMITS 6
5.4 INCORPORATION OF RESULTS INTO THE ROUTE ACCEPTANCE SAFETY CASE 7

6 APPENDIX A 8

LIST OF METHODOLOGY DOCUMENTS 8

1 Purpose

This Code of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

2 Scope

This Code of Practice is recommended for use by authors of Route Acceptance Safety Cases in accordance with Railway Group Standard GO/RT 3270 Route Acceptance of Rail Vehicles. It is compatible with the guidance and recommendations given in 'Yellow Book 3'.

The methods described in this Code of Practice provide a means of satisfying the requirements of Railway Group Standard GE/RT8015.

This Code of Practice covers Railtrack Controlled Infrastructure routes which are electrified on the 25kV AC overhead system or the 750V DC conductor rail system or which are dual-electrified, and the boundaries to these routes (including non-electrified boundaries).

This Code of Practice covers electrical compatibility from the point of view of interference with the safe operation of EE&CS equipment. It does not cover interference with the satisfactory operation of equipment in cases where there are no safety implications.

This Code of Practice is one of a suite of Codes of Practice (listed in Appendix A) which, when taken together and used in conjunction with a suitable asset register for the routes under consideration, constitute the tools necessary to demonstrate that a particular electric train can safely operate on Railtrack Controlled Infrastructure.

3 Definitions & Abbreviations

EE&CS	Electrical Engineering & Control Systems
Hazard	Any situation that could contribute to an accident.
ISA	Independent Safety Assessor
RASC	Route Acceptance Safety Case
RAR	Railtrack Asset Register
Route	The particular section of infrastructure for which acceptance of the rolling stock is being sought
Yellow Book 3	Engineering Safety Management Issue 3

4 Principle

The 'Methodology' described in this Code of Practice comprises a four-step process, which can be summarised as follows:

- identify the details of the items of EE&CS equipment installed on the route under consideration;
- obtain, from the Codes of Practice within this suite, the relevant electrical susceptibility characteristics for each item or group of items;
- compare the potential interference characteristics of the train with the various limits, or other requirements, contained in the Codes of Practice;
- incorporate the results into a Route Acceptance Safety Case for presentation to the acceptance authority.

These four steps are described in more detail in sections 5.1 - 5.4 below.

5 Process

5.1 Identification of Equipment on the Route

Details of the EE&CS equipment installed on the route and its boundaries must be obtained. This will typically include:

- types of track circuits;
- types of signalling control and interlocking systems;
- types of remote control systems;
- types of telecomms systems;
- types of lineside equipment including signals, points and lineside circuits;
- type of electrification system.

The preferred method of obtaining this information is for the RASC authors (or their representative) to ask the relevant Railtrack Acceptance Manager for the details of all the types of EE&CS equipment on the required routes and boundaries. Railtrack shall provide the necessary information, either from RAR or from an equivalent source such as Railtrack Record Group Offices.

Initially, it is only necessary to obtain a list of types of equipment installed on the route and boundaries, in order to decide which of the 'Methodology for the Demonstration of Compatibility' documents are relevant to the route.

5.2 Obtaining the susceptibility characteristics

For each generic type of equipment which has been identified on the route, the corresponding susceptibility characteristics can be found in the relevant 'Methodology' document, as listed in Appendix A. In each case, the extent to which detailed parameters of the installed equipment (i.e. over and above the simple list of equipment types referred to in 5.1) needs to be obtained can be determined by reference to the 'Methodology' document.

Parameters which are critical to the RASC need to be underwritten by the relevant Zonal Engineer. These parameters shall be identified by the Project, and the Acceptance Manager shall submit them to the Zone Engineer for underwriting.

Each 'Methodology' document gives information on the susceptibility parameters of the equipment concerned; these will generally vary according to the values of the detailed physical parameters of the installed equipment. In some cases, a simple 'pass/fail' benchmark level of tolerable interference is given; in other cases, more complex methodology such as the use of simulation models, or the use of recordings of interference from the train, is prescribed.

5.3 Comparison of train interference with susceptibility limits

The electrical interference which the rolling stock can potentially produce must be characterised. This needs to take account of normal steady state and transient operating conditions, and credible degraded mode and fault conditions.
Guidance on this specific activity is contained in document RT/E/C /50019 'Guidance on Traction & Auxiliary Monitoring'.

These characteristics are then compared with the susceptibility characteristics contained in the relevant 'Methodology' documents.

Where the train interference can be shown to be within a simple 'pass/fail' limit appropriate to the particular equipment in question in a robust manner, no further demonstration of compatibility is necessary as far as that equipment is concerned.

Where this cannot be shown, or where the nature of the infrastructure equipment is such that no such simple 'pass/fail' criteria can be specified, it will be necessary to carry out further steps, as described in the relevant Methodology document, to produce a satisfactory basis of compatibility. These further steps may require more detailed parameters of the installed equipment to be obtained.

The steps outlined above must be repeated in respect of each of the 'Methodology' documents which have been identified as relevant to the route, so that the compatibility of the rolling stock with all the EE&CS equipment on the route is systematically examined.

The levels in the documents take account of fleet operation. However, further analysis may be required if trains consistently generate interference levels which are near to the limit, and likely to be additive in nature, in respect of those locations (eg major stations and depots) where large numbers of trains are concentrated.

5.4 Incorporation of results into the Route Acceptance Safety Case

The results of using the 'Methodology' should be incorporated into the RASC by reference to the particular hazards which have been closed out by means of the 'Methodology'.

For example, where a hazard such as 'electrical interference from the train may cause wrong-side failure of DC track circuits' has been closed out by use of the 'Methodology' contained in RT/E/C/50004, this should be explained in the RASC.

6 APPENDIX A

LIST OF METHODOLOGY DOCUMENTS

RT/E/C/50001	Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure. (Guidance on how to use the compatibility documents)
RT/E/C/50002	Methodology for the demonstration of compatibility with Reed Track Circuits on the AC railway
RT/E/C/50003	Methodology for the demonstration of compatibility with Reed Track Circuits on the DC railway
RT/E/C/50004	Methodology for the demonstration of compatibility with DC (AC-immune) Track Circuits
RT/E/C/50005	Methodology for the demonstration of compatibility with 50Hz Single Rail Track Circuits
RT/E/C/50006	Methodology for the demonstration of compatibility with 50 Hz Double Rail Track Circuits
RT/E/C/50007	Methodology for the demonstration of compatibility with HVI Track Circuits
RT/E/C/50008	Methodology for the demonstration of compatibility with TI21 Track Circuits
RT/E/C/50009	Methodology for the demonstration of compatibility with FS2600 Track Circuits
RT/E/C/50010	Methodology for the demonstration of compatibility with Non-immune Track Circuits (Aster, DC(NI), Lucas)
RT/E/C/50011	Methodology for the demonstration of compatibility with Axle Counters
RT/E/C/50012	Methodology for the demonstration of compatibility with TPWS
RT/E/C/50013	Methodology for the demonstration of compatibility with Interlockings

RT/E/C/50014	Methodology for the demonstration of compatibility with Lineside Equipment on AC and DC Railways
RT/E/C/50015	Methodology for the demonstration of compatibility with Reed FDM Systems
RT/E/C/50016	Methodology for the demonstration of compatibility with Telecomms Systems
RT/E/C/50017	Methodology for the demonstration of compatibility with Traction Power Supplies & Systems
RT/E/C/50018	Methodology for the determination of interaction with Neighbouring Railways
RT/E/C/50019	Guidance on Traction & Auxiliary Monitoring

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019 **Issue 1**

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
Technical & Safety Advisor